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CHONGTIAN

Shenzhen Chongtian Optoelectronics Technology Co., Ltd.

SHENZHEN CHONGTIAN OPTOELECTRONIC TECHNOLIGY CO.,LTD

Product Specifications

SPECIFICATION

Product Name **Product Name**	0402 Red LED Surface Mount Light Emitting Diode	Customer Name **Customer**	
Product Model **Type**	**CT-1005URC-P4**	Customer part number **Customer P N**	
version number **Version**	A2	Date **Date**	2026/01/23

Energy conservation, emission reduction, outstanding innovation

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Chongtian Optoelectronics's Acknowledgment Section			Customer Approval Section	
CTGD Approval			Customer Approval	
Formulate **DRAW**	Review **CHECK**	Approval **APPROVE**	confirm **CONFIRM**	Approval **APPROVE**

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SMD

CT-1005URC-P4

Features:

ÿ 0402 package

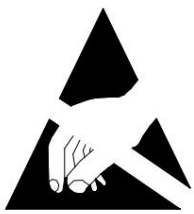
ÿ Top view LED

Packed on an 8mm carrier tape 7-inch reel.

Compatible with infrared and vapor phase reflow soldering processes

Lead-free

ÿ Compliant with RoHS[S]



ATTENTION

OBSERVEPRECAUTIONS
ELECTROSTATIC
SENSITIVEDEVICES

ÿ Note:

ÿ External dimensions (L/W/H) : 1.0 x 0.5 x 0.4 mm

Color: Red light

ÿ Colloid: Transparent colloid

EIA Standard Packaging

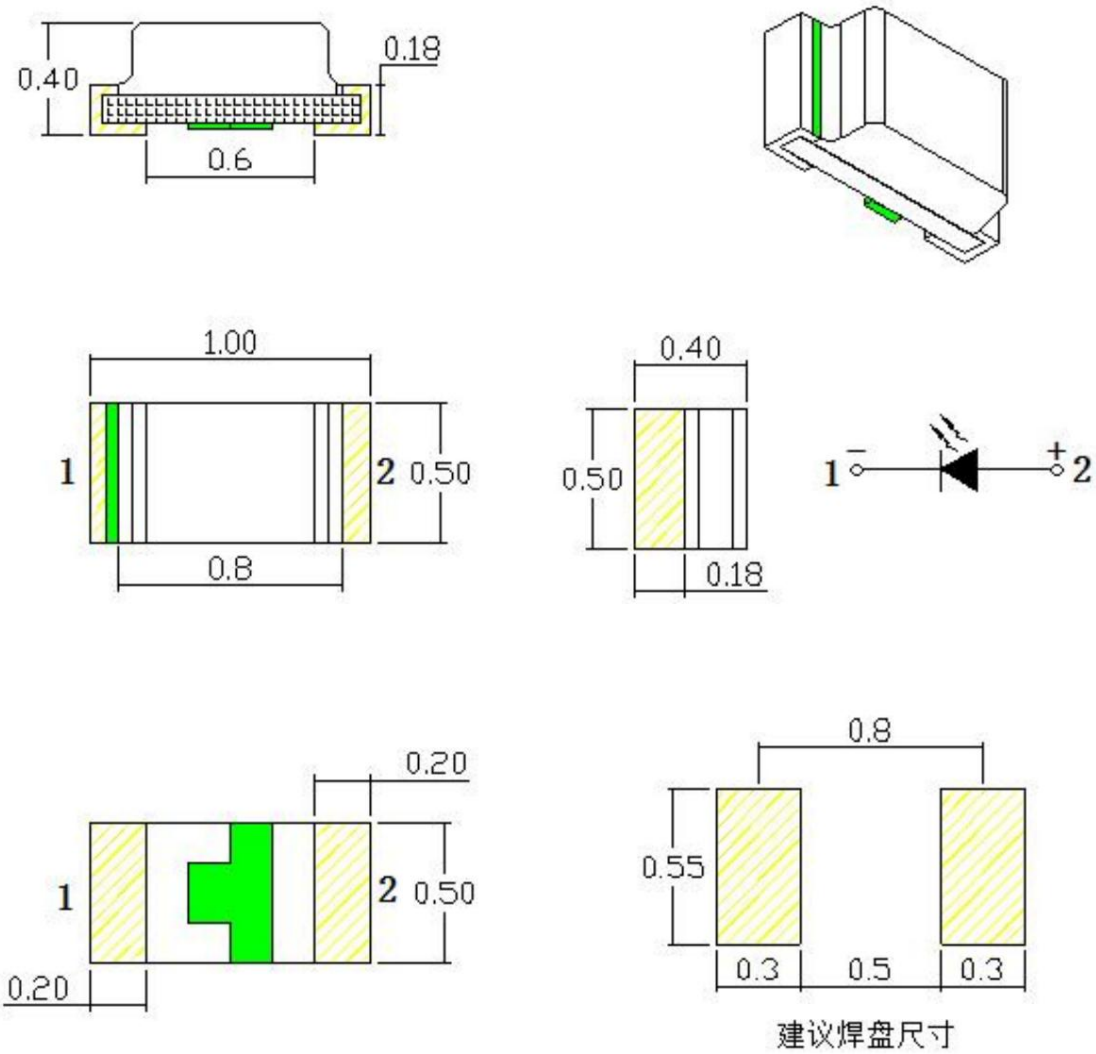
Environmentally friendly products that comply with RoHS requirements.

Suitable for automatic placement machines

Suitable for infrared reflow soldering processes.

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ÿ External dimensions and recommended pad dimensions:



Notes: 1. Unit: millimeters (mm) 2.

Tolerance: ± 0.10 mm unless otherwise specified

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ÿ Maximum absolute rating (Ta=25ÿ):

parameter	symbol	Maximum rating	unit
Power consumption	Pd	42	mW
Maximum pulse current (1/10 duty cycle, 0.1ms pulse width)	IFP	50	m.a.
Maximum forward DC operating current	IF	20	m.a.
Reverse voltage	VR	5	**In**
Operating ambient temperature	Topr	-40°C ~ + 85°C	
Storage ambient temperature	Test	-40°C ~ + 90°C	
Welding conditions	Tsol	Reflow soldering: 255±5°C, 10s Manual soldering: 300°C, 3 seconds	

ÿ: Photoelectric parameters (Ta=25ÿ)

parameter	symbol	Minimum value represents the maximum value, and the unit test condition is as follows.				
Light intensity	IV	100	--	250	mcd	IF =20mA
Half-intensity angle 2ÿ1/2		--	120	--	you	IF =20mA
dominant wavelength	ÿd	620	--	630	nm	IF =20mA
Forward voltage VF		1.8	--	2.2	In	IF =20mA
Reverse current IR		--	--	1	a	VR=5V

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ÿ Brightness levels:

code	Minimum value	Maximum value	unit	Test conditions
M1	100	150	mcd	IF=20mA
M2	150	200		
M3	200	250		

Note: Light intensity error ± 11%

Wavelength grading:

code	Minimum value	Maximum value	unit	Test conditions
N1	620	625	nm	IF=20mA
N2	625	630		

Note: Wavelength error ± 1 nm

Voltage range:

code	Minimum value	Maximum value	unit	Test conditions
V1	1.8	2.0	In	IF=20mA
V2	2.0	2.2		

Note: Forward voltage error ± 0.02V

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ÿ Characteristic curves of representative electrical parameters:

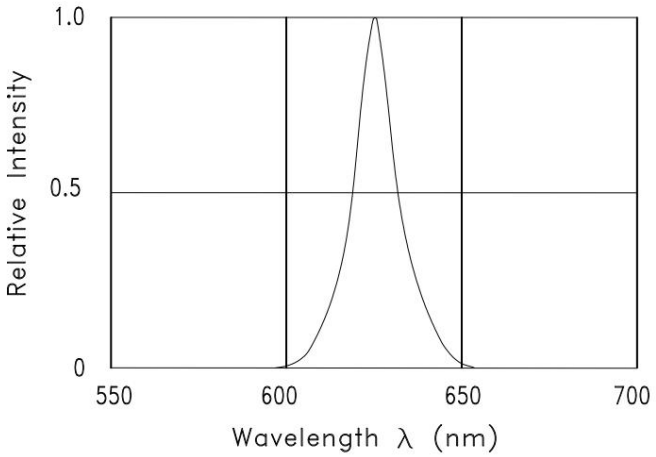


Fig.1 RELATIVE INTENSITY VS. WAVELENGTH

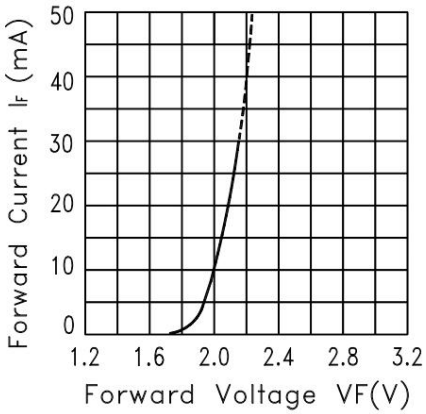


Fig.2 FORWARD CURRENT VS. FORWARD VOLTAGE

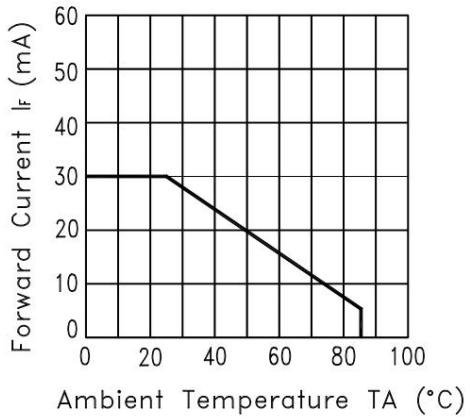


Fig.3 FORWARD CURRENT DERATING CURVE

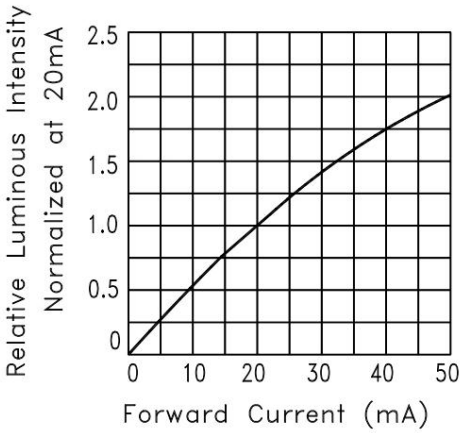


Fig.4 RELATIVE LUMINOUS INTENSITY VS. FORWARD CURRENT

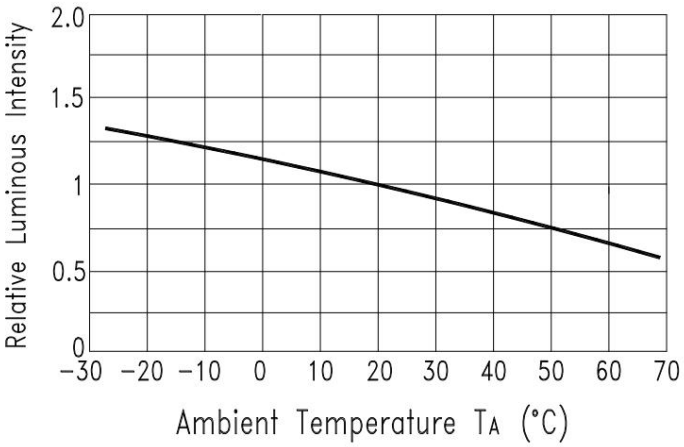


Fig.5 Luminous Intensity vs.Ambient Temperature

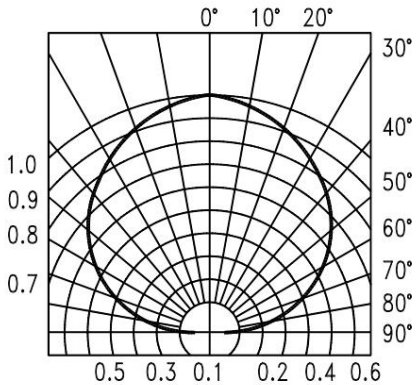


Fig.6 SPATIAL DISTRIBUTION

Note: Unless otherwise specified, the ambient temperature for testing is 25 ± 3 °C.C

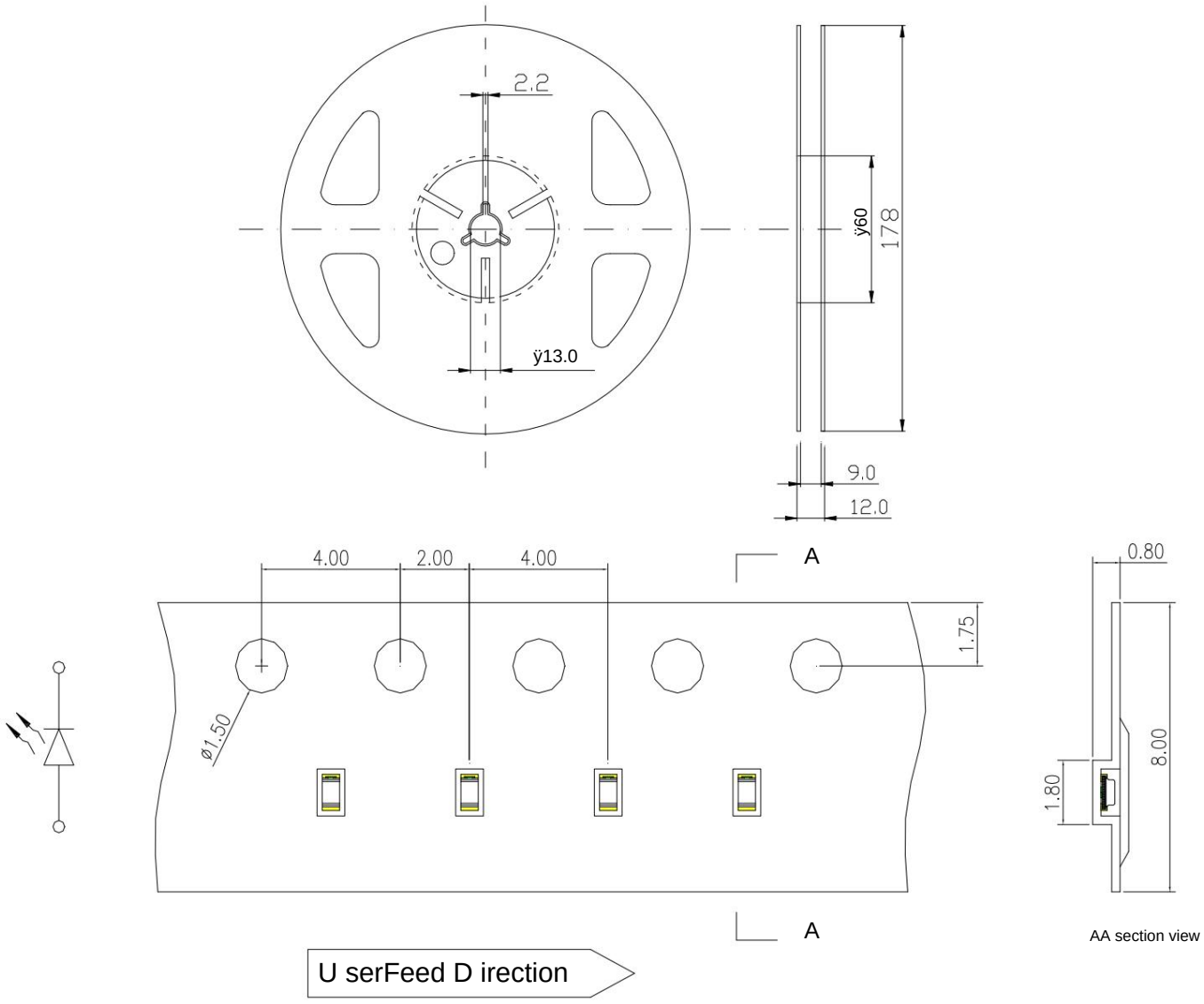
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ÿLabel identification:

CAT：光强 （单位（mcd））
HUE：波长 （单位（nm））
REF：电压 （单位（V））



Packaging carrier belt and disc dimensions:

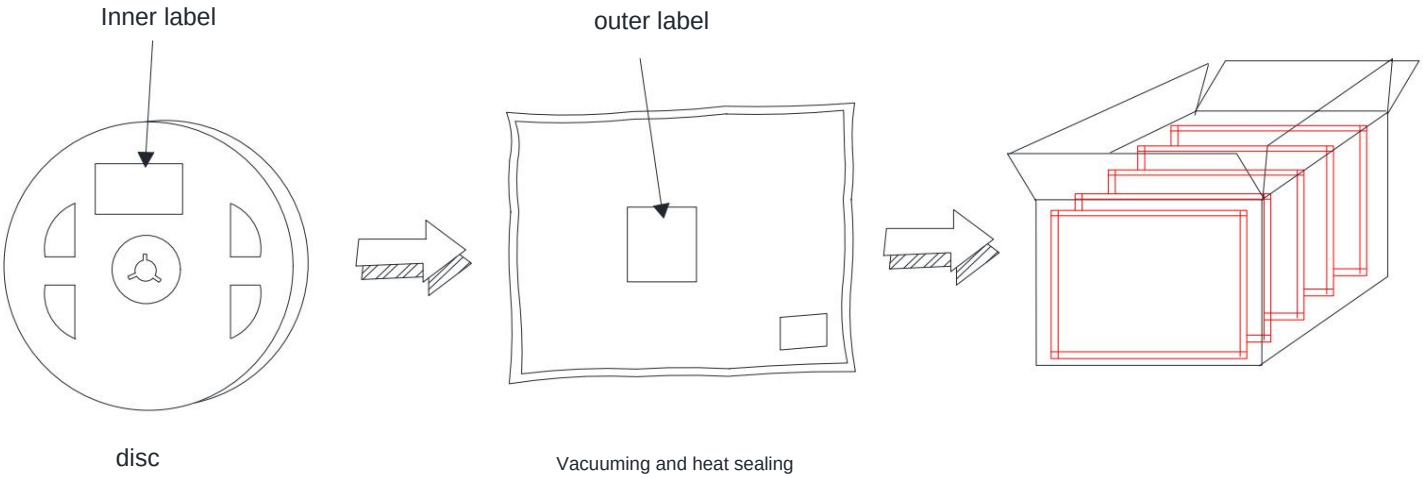


Notes: 1. Unit: millimeters (mm) 2.

Tolerance: ± 0.15 mm unless otherwise specified.

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Inner and outer packaging:



ÿ Trustworthiness Experiment:

Test Project	Test conditions	Test count reference standard		Failure determination standard	Number of failed LEDs **Quantity (PCS)**
Moisture resistance rating	1. Reflow soldering maximum temperature = 260ÿ, 10 seconds, 2 cycles Flow soldering; 2. Storage conditions before reflow soldering: 30°C, relative humidity Degree = 70%, 168H;		JEITA ED-4701 300.301	ÿ1	0/22
Welding reliability (Lead-free reflow) weld)	Maximum reflow soldering temperature = 255±5ÿ, 5 seconds (lead-free) (Reflow soldering)		JEITA ED-4701 303 303A	ÿ2	0/22
Hot and cold circulation	-40ÿ for 30 minutes ~ 25ÿ for 5 minutes 100ÿ for 30 minutes ~ 25ÿ for 5 minutes	300 cycle	JESD22-A104	ÿ1	0/22
thermal shock	-35ÿ for 15 minutes Conversion time: 3 minutes 85ÿ for 15 minutes	300 cycle	JESD22-A106	ÿ1	0/22
High temperature storage	Ta=100ÿ	1000 small hour	JESD22-A103	ÿ1	0/22
Low temperature storage	Ta=-40ÿ	1000 small hour	JESD22-A119	ÿ1	0/22
Aging at room temperature	Ta=25ÿ IF=20mA	1000 hours	JESD22-A108	ÿ1	0/22

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Precautions for use:

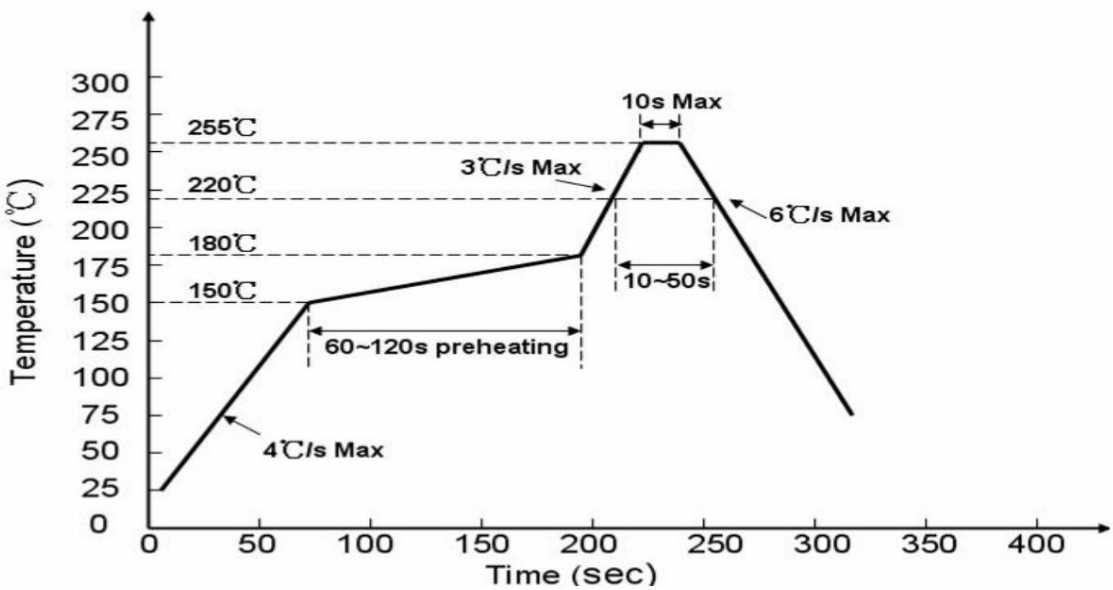
ÿ Usage:

1. Excessive temperature can affect the brightness and other performance of LEDs. Therefore, to ensure better performance of LEDs, they should be kept away from heat sources.

ÿ Welding

1. Manual Soldering Operation:

(1) The soldering iron used must be less than 25W, the soldering iron temperature must be kept below 350ÿ, and the soldering time must not exceed 3 seconds. (2) The soldering iron tip must not come into contact with the adhesive. (3) After the product is soldered, it must be cooled to a temperature below 40ÿ before packaging. 2. Reflow Soldering Operation:



(1) Please refer to the following standards for the temperature profile of reflow soldering: (2) Do not modify the soldered surface after soldering. If modifications are necessary, they must be made without damaging the product. (3) Reflow soldering should be completed in one time and cannot be done in multiple stages (to prevent the product from getting damp during the process). (4) After soldering, do not package immediately. Allow it to cool naturally before packaging.

Storage :

1. When the original packaging is unopened, the recommended storage environment is: temperature 5ÿÿ30ÿ, humidity below 60%RH. When inventory exceeds two... It lasts for one month. Dehumidification should be performed before use, under conditions of 60ÿ/24 hours.

2. After opening the original packaging, the recommended storage environment is: temperature 5-30°C, humidity below 60%;

3. LEDs are humidity-sensitive components. To prevent moisture absorption, it is recommended to store them in a sealed container with desiccant after opening the packaging, or 4. Store in a nitrogen-filled dehumidifier; 5. After opening the packaging, the components should be used within 168 hours (7 days); and soldering should be completed as soon as possible after placement; 6. If the desiccant is ineffective or the components are exposed to air for more than 168 hours (7 days), dehumidification treatment should be performed; Baking conditions: 60ÿ/24 Hour.

6. To avoid losses caused by materials absorbing moisture during the customer's production process, please strictly adhere to the above requirements.